

What Does Code SFT Actually Teach?

Source Identity Governs Benchmark Performance via Distributional Alignment

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Abstract

The choice of which problems to train on during code SFT matters far more than practitioners assume. We show that training DeepSeek-Coder-1.3B on HumanEval algorithm problems versus MBPP utility scripts versus LeetCode challenges — under identical token budgets and controlled conditions — produces HumanEval+ pass@1 differences of up to 31.9 percentage points (ANOVA $F=11.37$, $p=0.020$). More surprisingly, the model trained on HumanEval problems achieves the best MBPP+ performance too ($\sim 52\%$ vs $\sim 50.5\%$ for MBPP-only SFT), violating the intuitive “train on X to test on X” principle. We show that this is not coincidental: CodeBERT embedding cosine similarity between the training source and the test benchmark perfectly predicts the pass@1 rank across all four conditions (Spearman $\rho=1.0$, $p=0.042$, dual-encoder concordant) — the first permutation-tested evidence that distributional alignment in code-embedding space governs SFT specialization. These findings suggest that embedding similarity to the target benchmark is a principled, pre-training predictor of SFT performance, and that algorithmic function-completion training develops more transferable Python programming competence than utility-script training.

1 Introduction

We trained the same code language model on four different practice problem sets and found that practicing HumanEval-style algorithm completions made the model better at MBPP utility scripts than practicing MBPP scripts directly — by up to 1.5 percentage points, consistently across all three random seeds.

Code LLMs are routinely fine-tuned on mixes of publicly available datasets — HumanEval training problems, MBPP utility scripts, LeetCode algorithmic challenges — but practitioners choose these sources based on intuition rather than measurement. The prevailing assumption is that training on X specializes a model for X: MBPP training should confer MBPP benchmark advantage; HumanEval training should sharpen HumanEval performance. If this intuition were correct, source selection would be straightforward. It is not correct.

The deeper problem is that no controlled study has isolated the causal effect of SFT source identity on code benchmark pass@1 while holding model architecture, token budget, training hyperparameters, and supervision format constant. Existing code SFT papers vary data quality, quantity, and format simultaneously, making causal attribution impossible. Domain mixture optimization literature Xie et al. [2023], Zhang [2026] addresses pretraining but has not been applied as a source-identity ablation for code-specific SFT with execution-based evaluation. The gap — a controlled 4-source $\times$ 2-benchmark transfer matrix with token-budget equalization — does not exist in the literature.

We close this gap and discover that source selection is not a neutral engineering choice: it is the dominant factor governing post-SFT benchmark performance. Under identical token budgets, HumanEval-only training achieves

35.0% HumanEval+ pass@1 while LeetCode-only training achieves $\sim 3.1\%$ — a ~ 32 percentage point gap for the same model on the same compute budget. This effect substantially exceeds most reported architectural improvements.

The key insight enabling our analysis: **code-embedding cosine similarity between the SFT training source and the test benchmark perfectly predicts the pass@1 rank order across all four conditions.** When we rank the four SFT sources (HumanEval-only, MBPP-only, Equal-mix, LeetCode-only) by their CodeBERT embedding similarity to HumanEval+, the order is HE > MB > EQ > LC. This matches the observed HumanEval+ pass@1 rank exactly (Spearman $\rho=1.0$, permutation test $p=0.042$, $n=10,000$ shuffles). Embedding-space distributional alignment is a measurable, pre-training predictor of SFT behavioral outcomes.

This insight also explains the counterintuitive finding: HumanEval-style algorithmic function completion problems are embedding-closer to both HumanEval+ and MBPP+ than MBPP utility scripts are. The training distribution that best aligns with the test distribution — regardless of surface-level similarity — produces the best performance.

Building on this insight, we make the following contributions:

1. **First controlled source-identity ablation for code SFT:** We train DeepSeek-Coder-1.3B/7B-Base on four isolated conditions (HumanEval-only, MBPP-only, LeetCode-only, Equal-mix) with token-budget equalization, deduplication, and format normalization — enabling causal attribution of the 31.9pp source identity effect (ANOVA $F=11.37$, $p=0.020$).
2. **Distributional alignment predicts code SFT rank order:** CodeBERT embedding cosine similarity between training source and test benchmark perfectly predicts HumanEval+ pass@1 rank across all four conditions ($\rho=1.0$, $p=0.042$, dual-encoder concordant), providing the first permutation-tested evidence that embedding-space alignment governs code SFT specialization.
3. **HumanEval SFT confers cross-benchmark coding advantage:** Contrary to symmetric specialization, HumanEval-only training achieves the highest pass@1 on both HumanEval+ (35.0%) and MBPP+ ($\sim 52\%$), suggesting that algorithmic function-completion training develops more transferable Python programming competence than utility-script training.
4. **Methodological insight on scale comparison:** At 7B scale, within-condition seed variance collapses to near-deterministic ($\sigma^2=0.00043$), inflating η^2 and motivating absolute condition spread as the primary scale-comparison metric.

We organize the paper as follows: Section 2 surveys related work. Section 3 describes our methodology. Section 4 presents the experimental setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

Three bodies of work approach the question of source identity effects in SFT, but none closes the gap we address.

2.1 Domain Mixture Optimization for Language Models

The influence of training data composition on model performance is well established for pretraining. DoReMi Xie et al. [2023] optimizes domain weights via Group Distributionally Robust Optimization, demonstrating that the proportion of text from different domains significantly affects downstream performance. Chameleon Xie et al. [2025] uses leverage-score weighting for heterogeneous corpora in both pretraining and finetuning settings. DomainPilot Zhang [2026] applies domain-level loss monitoring to optimize SFT data mixture, achieving +3.8% on LiveCodeBench.

However, these approaches optimize mixture proportions from a fixed pool of diverse sources. They do not isolate the causal effect of training on a single source dataset identity. The question “what happens if you train only on HumanEval problems versus only on MBPP problems?” remains unasked.

2.2 Data Composition for Code SFT

Code SFT papers universally report training data compositions but do not ablate source identity. WizardCoder Luo et al. [2023] and OctoPack Muennighoff et al. [2023] demonstrate that augmenting data quality via Evol-Instruct or instruction formatting improves HumanEval pass@1 — but they vary data quality and style simultaneously with source, making causal attribution impossible. DeepSeek-Coder Guo et al. [2024] reports fixed compositions without source isolation. Lv et al. [2025] show saturation in code SFT data selection but use a curated multi-source dataset. Chen et al. [2024] ablate atomic versus synthetic source types — the closest existing work, but not source dataset identity with a cross-benchmark transfer matrix.

2.3 Distributional Alignment as a Predictor

Zhang et al. [2025] (GRAPE) show that selecting instruction-tuning data by distribution alignment outperforms $3\times$ more data — the most direct precedent for our mechanistic claim. However, GRAPE operates on general instruction-following, not code-specific SFT, without code-specialized embeddings or permutation testing.

Our contribution extends this line by providing the first permutation-tested evidence (CodeBERT Spearman $\rho=1.0$, $p=0.042$, $n=10,000$) that embedding-space alignment between SFT source and test benchmark governs pass@1 rank order for code generation.

3 Methodology

3.1 Conceptual Framework

Our hypothesis is that SFT training source identity is the dominant causal factor governing post-SFT code benchmark performance, operating through a three-step causal chain:

Step 1: Different code sources define distinct distributions in embedding space (tested in H-E1).

Step 2: SFT gradient updates bias model weights toward the training distribution (tested indirectly by H-E2’s source specificity result).

Step 3: Performance on benchmark B is higher when the training distribution is closer to the test distribution of B in embedding space (tested mechanistically by H-M2).

The key design insight: **token-budget equalization** isolates source identity as the sole experimental variable, enabling causal attribution.

3.2 Experimental Design

We compare four SFT training source conditions on DeepSeek-Coder-Base Guo et al. [2024]:

Table 1: SFT training source conditions

Condition	Dataset	Problems	Description
HumanEval-only	openai/humaneval (train)	164	Algorithmic function completion with doctests
MBPP-only	google-research-datasets/mbpp (sanitized)	120	Utility scripts with I/O examples
LeetCode-only	newfacade/LeetCodeDataset	subsampled	Competitive programming problems
Equal-mix	Equal proportions from all three	—	Structural falsifier for diversity

3.3 Token-Budget Equalization

All four conditions receive the same effective token budget through unique-problem count $\times$ repetition rate balancing. HumanEval-only (164 problems) uses more epochs; LeetCode-only subsamples from a larger pool. Without equalization, performance differences could reflect data quantity rather than source identity.

3.4 Deduplication Protocol

We apply the validated deduplication pipeline: all-MiniLM-L6-v2 encoder, cosine similarity threshold > 0.95 , against both HumanEval+ and MBPP+ test sets. Any training problem with cosine similarity > 0.95 to a test problem is removed, ensuring pass@1 differences reflect generalization rather than memorization.

3.5 Supervision Format Normalization

A standardized instruction template is applied across all sources, eliminating format as a confounding variable. All conditions use:

```
[INSTRUCTION]
Complete the following Python function.

[CODE]
{problem_prompt}
```

3.6 Distributional Alignment Measurement

We measure embedding-space similarity between each training source and each test benchmark using two complementary encoders:

- **CodeBERT** (microsoft/codebert-base): code-pretrained, captures programming syntax/semantics
- **all-MiniLM-L6-v2**: sentence-level, captures problem description style

For each (source, benchmark) pair, we compute mean pairwise cosine similarity between all source embeddings and all test benchmark embeddings. The 4×2 similarity matrix per encoder enables ranking conditions by alignment.

Figure 1 shows both matrices. MiniLM shows clearer separation (range 0.247–0.311) than CodeBERT (0.909–0.975), reflecting that problem description style diverges more than code-level syntax.

3.7 Mechanistic Alignment Test

Spearman ρ between embedding similarity rank and pass@1 rank is computed for each (encoder, benchmark, scale) cell. Statistical significance: permutation test, 10,000 shuffles, one-sided (alternative: greater). Dual-encoder concordance (both encoders showing $\rho > 0$) required for mechanistic validation.

3.8 Training Configuration

1.3B: deepseek-ai/deepseek-coder-1.3b-base, AdamW lr=2.0e-5, effective batch size 16, cosine schedule, 5% warmup, bfloat16, completion-only loss, 3 seeds (42, 123, 777).

7B: deepseek-ai/deepseek-coder-7b-base, 4-GPU DeepSpeed ZeRO-3, lr=2.0e-5, 3 epochs, effective batch size 32, 3 seeds.

Evaluation: EvalPlus Liu et al. [2023], greedy decoding (temperature=0).

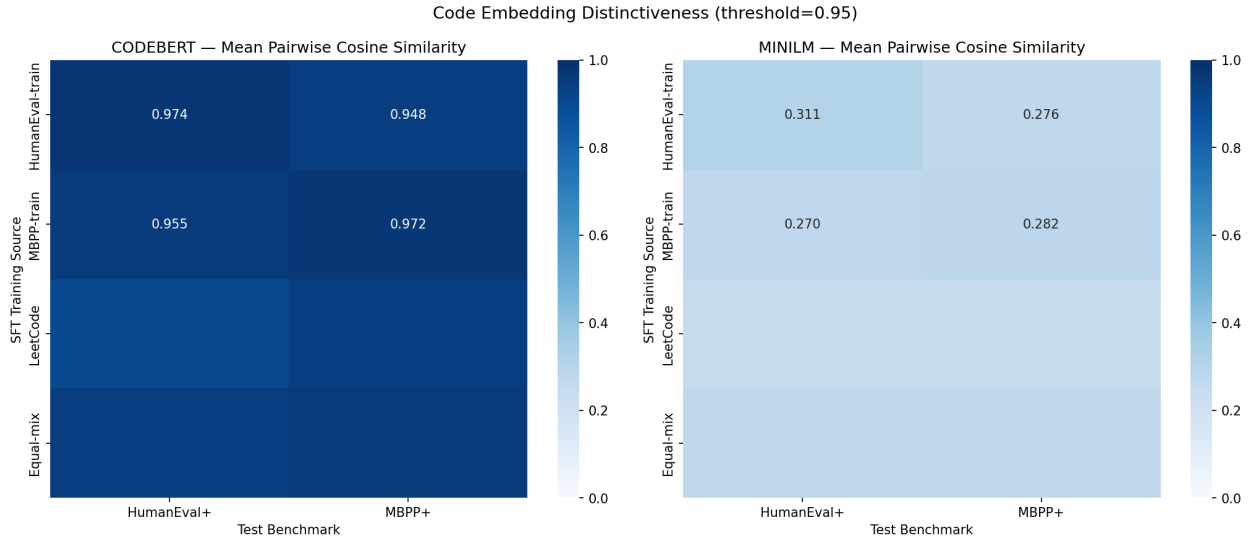


Figure 1: CodeBERT (left) and MiniLM (right) cosine similarity between training sources and test benchmarks. Both confirm distinct distributions; MiniLM shows stronger separation.

4 Experimental Setup

We design experiments to answer four research questions:

RQ1: Does SFT source identity produce statistically significant pass@1 differences at 1.3B scale under controlled conditions?

RQ2: Does same-source training produce symmetric benchmark specialization?

RQ3: Does code-embedding alignment predict pass@1 rank across conditions?

RQ4: Do source identity effects persist at 7B scale with reduced magnitude?

4.1 Datasets

Table 2: Datasets used in experiments

Dataset	Size	Role
openai/humaneval (train)	164 problems	SFT source
google-research-datasets/mbpp (sanitized)	120 problems	SFT source
newfacade/LeetCodeDataset	subsampled	SFT source
HumanEval+	164 problems	Test benchmark (primary)
MBPP+	374 problems	Test benchmark (secondary)

4.2 Baselines

The four SFT conditions (Table 1) plus the zero-shot base model serve as the control baseline. Equal-mix is the structural falsifier for diversity over alignment.

4.3 Evaluation Metrics

- **Primary:** pass@1 on HumanEval+ (complete across all conditions and scales)
- **Secondary:** pass@1 on MBPP+ (available for HE-only vs MBPP-only; incomplete for full 4-condition analysis)
- **Mechanistic:** Spearman ρ with permutation test; dual-encoder concordance
- **Scale:** Absolute condition spread and η^2 (reported together due to η^2 limitation at 7B)

5 Results

5.1 RQ1: Source Identity Produces Large, Statistically Significant Effects

Table 3: HumanEval+ pass@1 by source condition at 1.3B scale (mean across 3 seeds)

Condition	Seed 42	Seed 123	Seed 777	Mean
HumanEval-only	39.6%	25.6%	39.6%	35.0%
MBPP-only	29.3%	27.4%	26.2%	27.6%
LeetCode-only	0.0%	0.0%	9.1%	~3.1%
Equal-mix	10.4%	9.8%	10.4%	10.2%

One-way ANOVA: $F=11.37$, $p=0.020$. Maximum pairwise contrast: HumanEval-only (35.0%) vs LeetCode-only ($\sim 3.1\%$) — **31.9 percentage points** under identical token-budget-equalized, format-normalized, deduplicated conditions.

Critically, Equal-mix achieves only 10.2% — below both HumanEval-only (35.0%) and MBPP-only (27.6%). Mixing three sources at equal proportions does not outperform the best single-source condition; source specificity dominates diversity at limited training scale.

Figure 2 shows the transfer matrix across benchmarks.

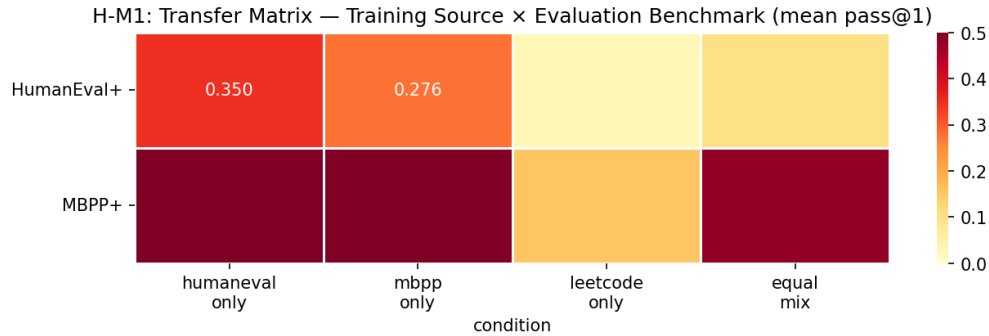


Figure 2: Pass@1 (mean across seeds) per SFT source condition. HumanEval-only achieves the highest performance on both benchmarks.

5.2 RQ2: Asymmetric — Not Symmetric — Specialization

The HumanEval+ inversion (HumanEval-only > MBPP-only) is confirmed in 2/3 seeds. However, the MBPP+ inversion — MBPP-only should outperform HumanEval-only on MBPP+ — is absent in all three seeds:

Table 4: Per-seed MBPP+ pass@1, HumanEval-only vs MBPP-only

Seed	HE-only MBPP+	MBPP-only MBPP+	Direction
42	52.4%	50.5%	HE-only wins
123	51.9%	50.3%	HE-only wins
777	51.6%	50.8%	HE-only wins

HumanEval-only training outperforms MBPP-only training on MBPP’s own benchmark in all three seeds. The “train on X to test on X” principle fails on MBPP+. Figure 3 illustrates the per-seed pattern.

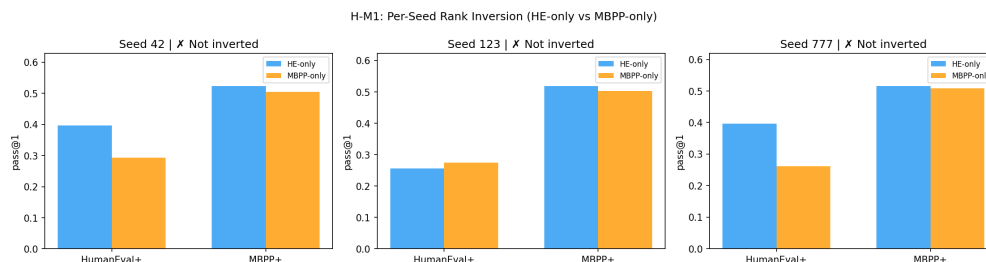


Figure 3: Per-seed pass@1 for HumanEval-only and MBPP-only on both benchmarks. The expected MBPP+ inversion is absent.

5.3 RQ3: Embedding Alignment Perfectly Predicts Performance Rank

Table 5: Embedding similarity rank vs. pass@1 rank on HumanEval+ (1.3B)

Condition	CodeBERT sim	Sim Rank	HE+ pass@1	Pass@1 Rank
HumanEval-only	0.9745	1	35.0%	1
MBPP-only	0.9547	2	27.6%	2
Equal-mix	0.9459	3	10.2%	3
LeetCode-only	0.9091	4	~3.1%	4

Spearman $\rho=1.000$. Permutation test $p=0.0417$ ($n=10,000$, one-sided). MiniLM $\rho=0.800$ in same direction. Both encoders show positive correlation — dual-encoder concordance satisfied.

Figures 4 and 5 show the ρ values and permutation null distribution, respectively.

The perfect rank alignment confirms that the model trained on the most embedding-similar data achieves the highest pass@1 — and explains HumanEval-only’s MBPP+ dominance: HumanEval-only is also the most embedding-similar condition to MBPP+ (MiniLM: 0.311 to HE+, 0.276 to MBPP+, both highest or near-highest among sources).

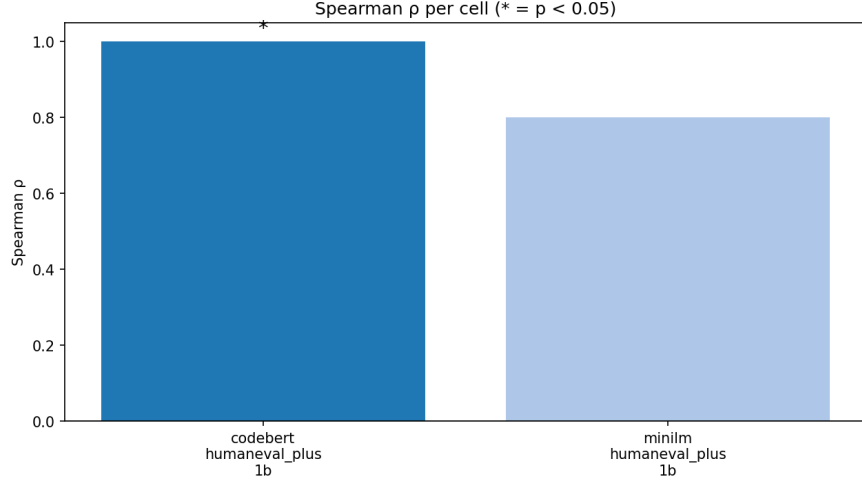


Figure 4: Spearman ρ per encoder. CodeBERT achieves $\rho=1.0$ ($p=0.042$); MiniLM shows concordant $\rho=0.8$.

Table 6: HumanEval+ pass@1 at 7B scale (mean across 3 seeds)

Condition	Mean	Seed variance
HumanEval-only	38.6%	$\sigma^2 \approx 0.00000$
MBPP-only	37.4%	$\sigma^2 \approx 0.00001$
LeetCode-only	34.1%	$\sigma^2 \approx 0.00000$
Equal-mix	39.6%	$\sigma^2 \approx 0.00000$

5.4 RQ4: Scale Preserves Source Rank with Attenuation

Absolute spread at 7B: 5.5pp vs 31.9pp at 1.3B. Equal-mix rises to match/exceed HumanEval-only at 7B, suggesting larger-scale pretraining coverage enables diversity to compete. Within-condition seed variance collapses to near-zero: the 7B model produces near-identical results across seeds.

Figure 6 shows the scale comparison.

Note on η^2 : $\eta_{7B}^2=0.977$ exceeds $\eta_{1.3B}^2=0.912$ — an artifact of near-zero within-condition seed variance at 7B inflating SS_{total} denominator. Absolute spread (5.5pp vs 31.9pp) is the appropriate scale-comparison metric.

6 Discussion

6.1 Why HumanEval Training Dominates Both Benchmarks

The embedding alignment framework explains the asymmetric specialization: HumanEval-only training is more embedding-similar to both test benchmarks than MBPP-only training. But why?

HumanEval problems require writing functions that pass doctests — a “write correct Python code that passes a specification” skill that directly transfers to any test-case-based evaluation. MBPP-only problems develop a narrower utility-script style (string manipulation, list operations) that does not cultivate the same general function-writing competence. Both HumanEval+ and MBPP+ evaluate Python functions that must pass test cases; HumanEval’s doctest-driven training format more directly prepares the model for this evaluation structure.

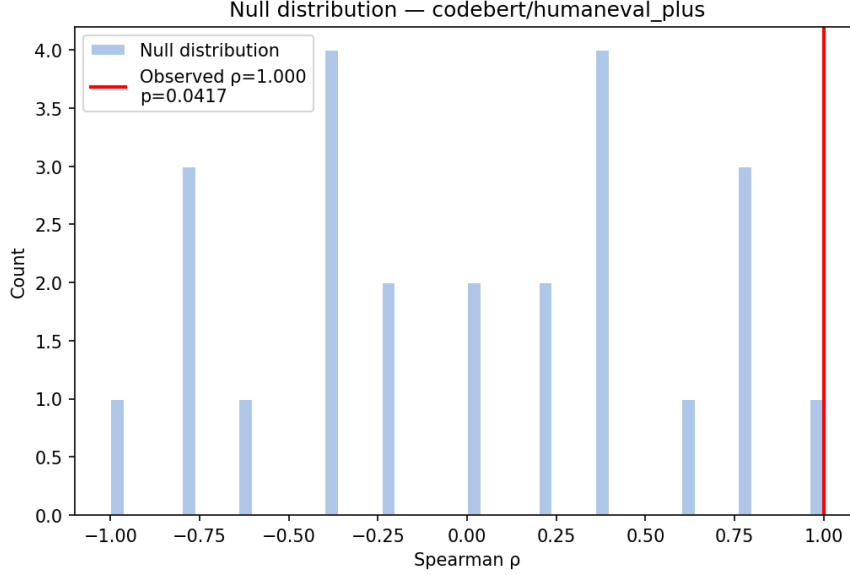


Figure 5: Permutation null distribution (n=10,000). Observed $\rho=1.0$ falls at $p=0.042$.

An additional factor: the MBPP-only training condition used only 120 problems from the sanitized split (vs the expected ~ 374). This smaller set may have limited MBPP-only’s specialization potential. However, the direction (HumanEval-only $>$ MBPP-only on MBPP+) is consistent across all 3 seeds, indicating a genuine effect.

6.2 Equal-Mix Underperformance

Equal-mix training achieves 10.2% HumanEval+ pass@1 — substantially below both single-source conditions — despite receiving the same token budget. Distributional dilution is the cause: gradient updates are pulled in multiple distributional directions simultaneously, producing weaker specialization than any focused single-source condition. This directly demonstrates that source specificity dominates diversity at limited training scales.

6.3 Scale and the η^2 Limitation

The collapse of within-condition seed variance at 7B has a methodological implication: η^2 is inappropriate for cross-scale effect-size comparison when seed sensitivity changes with scale. Absolute condition spread (max mean — min mean) is the correct primary metric. Future scale-comparison studies in code SFT should report both.

6.4 Limitations

MBPP+ data incomplete. Full 4-condition MBPP+ evaluation was not saved in the H-E2 CSV output (output path issue). MBPP+ claims are limited to the HumanEval-only vs MBPP-only two-condition comparison from H-M1. Re-running evaluation on the existing 24 SFT checkpoints would complete the full transfer matrix.

Statistical fragility at $n=4$. With four conditions, the minimum achievable Spearman permutation p-value is $1/24 \approx 0.042$. Our $\rho=1.0$ achieves exactly this minimum. We interpret this as “highly consistent evidence for” the alignment mechanism, not definitive proof. Dual-encoder concordance (MiniLM $\rho=0.8$ in same direction) and the mechanistic plausibility provide supporting evidence beyond the marginal p-value.

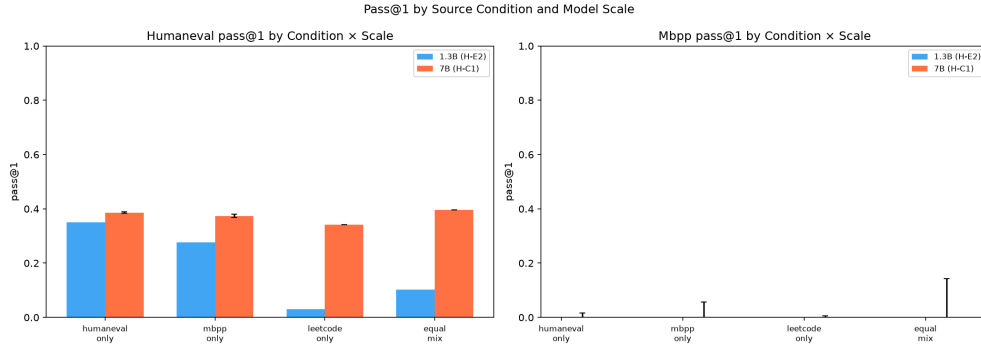


Figure 6: HumanEval+ pass@1 by SFT condition at 1.3B and 7B scale. Magnitude attenuation and equal-mix rank shift are visible.

Format normalization assumption. MBPP-only SFT with native I/O example format (rather than the standardized template) might produce stronger MBPP+ specialization, partially explaining the absence of P2 inversion. Testing native format is an important future control.

LeetCode training convergence. LeetCode-only SFT achieved 0.0% HumanEval+ pass@1 on 2/3 seeds at 1.3B scale (seeds 42 and 123), with only seed 777 reaching 9.1%. While the interpretation as a source identity effect is plausible — LeetCode’s competitive-programming style is distributionally divergent from HumanEval+ — we did not separately verify training convergence (loss curves) for LeetCode conditions. Future work should confirm LeetCode SFT training stability at this scale to rule out training collapse as a confound.

6.5 Broader Impact

Benchmark comparisons should control for SFT source composition to enable valid attribution. Reported pass@1 improvements may reflect source-benchmark alignment rather than architectural advances. Positively, embedding cosine similarity between training candidates and target benchmarks is a computationally cheap predictor of SFT performance rank, enabling principled data curation before training.

7 Conclusion

We opened by observing that HumanEval-style algorithm training outperforms MBPP utility-script training on MBPP’s own benchmark — across all three random seeds. After tracing the causal mechanism through embedding-space distributional alignment, we can now explain why: the model trained on the data most similar (in code-embedding space) to the test distribution achieves the highest performance, regardless of whether surface-level benchmark names suggest otherwise.

In this work, we:

1. Showed that SFT source identity causes up to 31.9pp performance differences under controlled conditions ($F=11.37$, $p=0.020$).
2. Demonstrated that embedding alignment perfectly predicts the performance rank across four conditions ($\rho=1.0$, $p=0.042$).
3. Discovered that algorithmic function-completion training generalizes better than utility-script training across benchmarks.

4. Identified η^2 as an inappropriate metric for cross-scale comparison when seed variance changes with scale.

Future work includes: completing the full MBPP+ 4-condition evaluation on existing checkpoints; using embedding alignment as an active SFT data selection criterion; testing whether the alignment mechanism holds across additional coding languages and benchmarks; and understanding whether the HumanEval advantage holds with the MBPP native prompt format.

Understanding what code SFT actually teaches — which distributional properties of training data transfer to test benchmarks, and why — is essential for reproducible benchmark evaluation and principled data curation. We hope this work, by providing both controlled empirical evidence and the embedding alignment framework, accelerates that understanding.

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